

Abhinav Raj

Integrated M.Sc. Computer Science (AI/ML)

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PROFESSIONAL SUMMARY

Integrated M.Sc. Computer Science (AI/ML) graduate candidate with a strong foundation in machine learning, deep learning, and biosignal processing. UGC-NET qualified (95.31 percentile, Assistant Professor & Ph.D. eligible) with hands-on experience in developing intelligent systems using TensorFlow, PyTorch, and Django. Experienced in real-time AI systems, deep learning research, and full-stack deployment, with interests in applied machine learning, automation, and research-driven development.

EDUCATION

Integrated M.Sc. in Computer Science with Specialization in AI and ML

Nehru Arts and Science College, Kanhangad | Kannur University | 2021–2026 (Result Awaiting)

- Relevant Coursework: Machine Learning, Deep Learning, Neural Networks, Computer Vision, NLP, Data Mining & Warehousing, Web Development, Database Systems, Software Engineering
- CGPA: 8.77 (based on [IX semesters](#))
- Completed all 10 semesters; final university result awaiting.

ACHIEVEMENTS

Qualified UGC-NET DECEMBER 2025 (Computer Science and Applications)

University Grants Commission (NTA) | [E-Certificate Number: 25D/03/031644](#) | Percentile: 95.31

PROJECTS

FrugalMyo: Real-Time Myoelectric Gesture Recognition using Deep Learning

Academic Project, Tenth Semester – 2026 |

Developed a real-time sEMG-based gesture recognition system using a custom EMG acquisition setup and deep learning.

- Designed and trained a 1D-CNN model (~87% accuracy) for hand gesture classification from raw EMG signals.
- Built an end-to-end pipeline for signal preprocessing, segmentation, training, and real-time inference.
- Integrated a Streamlit-based interface for live EMG visualization, gesture prediction, and confidence estimation.

Replication of NINAPRO DB1 Myoelectric Benchmark

Academic Project, Ninth Semester – 2025 | [GitHub: `github.com/ARJ010/ninapro-db1-benchmark-replication`](#)

Replicated the foundational 2015 Atzori et al. benchmark on the NINAPRO DB1 dataset to validate sEMG control pipelines.

- Engineered a custom two-pass numpy.memmap pipeline to process high-dimensional features on 8GB RAM
- Validated benchmark findings, identifying MAV + MLP as the optimal combination for real-time prosthetic control.
- Optimized classifier implementations, achieving significant training speed improvements.

Tech Stack: Python, NumPy, SciPy, scikit-learn

Checklist Management System (CMS) *Deployed & In Use at TH&Co Chartered Accountants*

Academic Project, Sixth Semester – 2023 | GitHub: github.com/ARJ010/checklist_app

Developed a role-based task management system to streamline procedural workflows for a CA firm.

- Implemented secure multi-role access (Admin, User, Checker) to manage audit workflows
- Enabled checklist automation, form tracking, and revision feedback loops

Tech Stack: Django, SQLite, HTML, CSS, JavaScript

Attendance Tracking System (ATS) *Deployed & In Use at Nehru Arts and Science College*

Independent Institutional Project – 2024 | GitHub: github.com/ARJ010/Attendance_Tracking_System

Developed an attendance management system for the Four-Year UG Programme at the institution.

- Built an instructor-friendly workflow for attendance logging and automated CSV report generation.
- Designed an optimized database schema for student, course, and attendance management.

Tech Stack: Django, MySQL, Bootstrap

SKILLS

- **Languages:** Python, C, Java, SQL, LaTeX, JavaScript, HTML, CSS
- **Libraries & Frameworks:** TensorFlow, PyTorch, scikit-learn, NumPy, SciPy, Pandas, Django, Streamlit
- **Tools & Platforms:** Git, GitHub, Jupyter Notebooks, MySQL, SQLite, VS Code
- **Development Areas:** Machine Learning, Deep Learning, Biosignal Processing (sEMG), Data Engineering, Web Development
- **Soft Skills:** Research Methodology, Technical Writing, Critical Thinking, Academic Communication, Problem-Solving

CERTIFICATIONS

TensorFlow Developer Professional Certificate

DeepLearning.AI – May 2025

Credential ID: UABTJPA585N8

- Completed full specialization covering CNNs, NLP, and Time Series Forecasting using TensorFlow 2.x
- Included hands-on projects and model implementation in real-world AI tasks

Mathematical Foundations for Machine Learning (Elite Certification)

NPTEL & IISc Bangalore – Oct 2025

Credential ID: NPTEL25CS136S959800072

- Awarded "Elite" status for performance in the 12-week advanced curriculum covering Linear Algebra, Probability, and Optimization

Project Completion Certificate – Checklist Management System

TH&Co Chartered Accountant Firm – Jul 2024

- Recognized for successful deployment and implementation of CMS

Web Development Basics (HTML, CSS, JavaScript)

University of Michigan – Nov 2023

- Credential IDs: F26NYDE6TBYC, 9NB9NJ6JV9L7, GGDU78ZXCLE7

Programming for Everybody (Getting Started with Python)

University of Michigan – Nov 2021

Credential ID: P2XYRZ3W3F96

- Covered foundational programming concepts in Python